

Imaging and Measurement

From light to a digital image

A short introduction

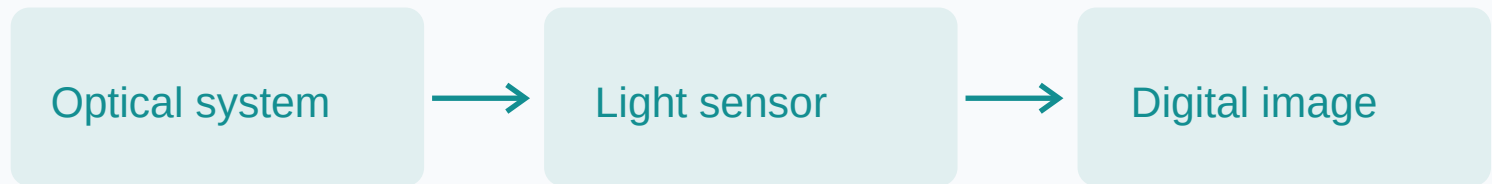


Course structure

Light enters the optical system.

The sensor converts light into an electrical signal.

A digital image is formed from sampled measurements.

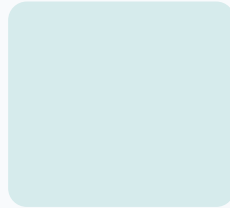


Optical sensor

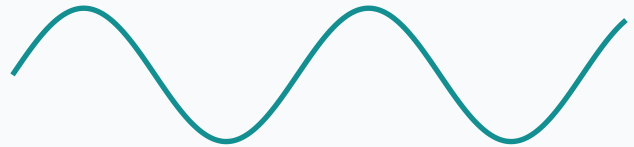
A sensor converts a physical input into a measurable signal.

An optical sensor responds to incoming light.

Incoming light



Electrical signal

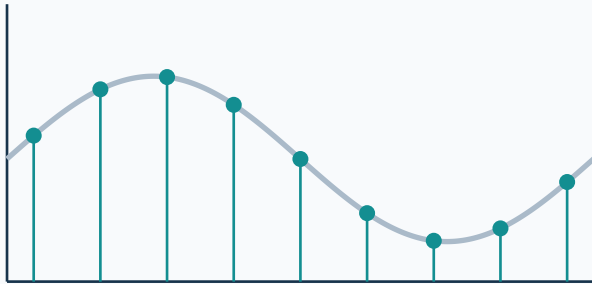


Sampling and quantization

Sampling selects values at discrete positions.

Quantization maps values to a finite set of levels.

Discrete positions



Finite levels

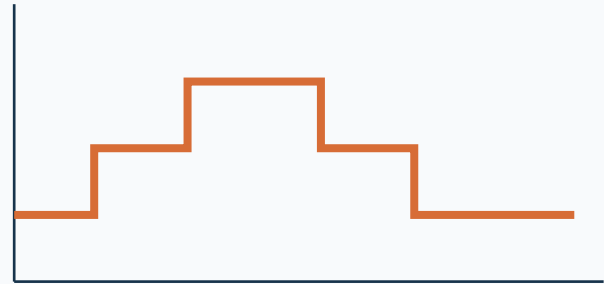


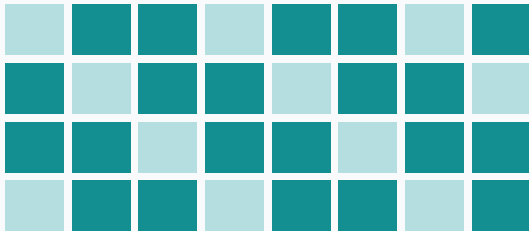
Image size and bit depth

Image size counts pixels along each dimension.

Bit depth sets the number of bits used per pixel.

For grayscale data, 8 bits represent 256 intensity levels.

Pixel grid



Intensity levels

